Coordination-Collapse Phase Boundary in Policy-Gradient Cooperative MARL

Anonymous submission

Abstract

Cooperative multi-agent reinforcement learning (MARL) can collapse into a degenerate *lazy-agent* policy when reward centralization is too low, a failure described only qualitatively. We make it measurable, estimating the critical centralization w_{crit} at which coordination gives way to collapse with a pre-registered definition and bootstrap intervals. Using this boundary as an instrument, we find that behavioral *loss aversion* ($\lambda > 1$) shifts it rightward—loss-averse agents demand more centralization to coordinate. This *direction* is the robust result: monotone in λ, sign-fixed a priori by a mean-field argument, and replicated across four environment–algorithm settings, as is the boundary's log-linear metastable slide with the horizon. The *magnitudes* are substrate-specific—the $w_{\text{crit}}(\lambda)$ law fitting the supply chain to RMSE 0.014 misfits Level-Based Foraging at 0.15, and curvature lowers w_{crit} here but raises it elsewhere. The instrument also exposes that prospect theory's two components move the boundary in *opposite* directions, and that a single *downstream* loss-averse agent can set a directed-flow system's requirement—a falsifiable prediction. We claim the mechanism at the sign-and-form level, and release the pre-registered protocol as a reusable instrument for measuring where any cooperative system collapses.

Introduction

A team of learning agents sharing a task can fall into a trap their designers never intended: instead of coordinating, every agent quietly defaults to inaction, and the team settles into a stable, useless equilibrium. Cooperative MARL calls this *coordination collapse*, with the *lazy-agent* policy as its endpoint (Liu et al. 2023; Shi and Peng 2023)—an instance of the sequential social dilemma that arises whenever private incentives drift from the collective good (Leibo et al. 2017). Whether a team falls in is governed by one design dial, the *reward centralization* $w \in [0, 1]$: the share of each agent's reward tied to the team's outcome rather than its own. Turn w down and an agent's private performance drowns out its contribution, so cooperating earns no learnable signal and the safe policy wins. The field knows this only as a direction—*more centralization helps* (Akella 2025; Liu et al. 2023)—and stops there. *Where*, in w, a working team tips into collapse has never been measured. The nearest work mitigates the lazy-agent pathology without locating a boundary (Liu et al. 2023; Shi and Peng 2023; Weng and Lee 2026), or sweeps neighbouring axes—credit-assignment level (Zhao and Xie 2025), reward weights (Jiang et al. 2020). Closest in spirit, Yamaguchi (2025) build a phase diagram for independent Q-learning over *environmental* axes (grid size, density), and Capannoli et al. (2026) prove a metastable boundary in network opinion dynamics under a fixed update rule; neither measures a reward-structure threshold in gradient-trained cooperative policies, with confidence intervals, that a behavioral bias then moves.

Pinning that number down matters now, because the agents being deployed are no longer neutral optimizers. In automated replenishment, fraud review, and energy dispatch, learning agents act for or beside people and absorb human preference structure by construction—whether trained on human choices through preference-aligned objectives that are themselves prospect-theoretic in form (Ethayarajh et al. 2024), or sharing the loop with human operators (Lalmohammed et al. 2025). The sturdiest such regularity is *loss aversion*: a loss looms larger than an equal gain (Kahneman and Tversky 1979; Tversky and Kahneman 1992a). Behavioral economics knows what this does to one chooser; it is silent on what it does to a *group* learning to coordinate (Leibo et al. 2017; Lowe et al. 2017). A measured collapse boundary is the instrument that question needs, and it extends a thin existing bridge between behavioral preferences and collective learning dynamics—inequity aversion in social dilemmas (Hughes et al. 2018), deterministic limits of multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019)—to the prospect-theoretic preferences that today's training pipelines instill.

We therefore build the ruler first, and treat it as the

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paper's central object. Following the posture physics takes toward a phase boundary, we fix a binary collapse criterion in advance, freeze it, fit $P(\text{collapse} \mid w)$, and bootstrap confidence intervals on w_{crit}. (We mean "phase boundary" empirically—a sharp, metastable threshold whose $T \to \infty$ limit is the boundary proper—not a thermodynamic singularity.) Two design choices keep the readings honest: we apply cumulative prospect theory (CPT) as a *reward transform* after blending rather than inside the update, so the optimizer is bit-identical across conditions and any movement of w_{crit} is behavioral; and we re-measure at several horizons, since a drifting threshold could pass for a constant. Our contributions are: (i) the first confidence-intervalled phase boundary of coordination collapse in deep cooperative MARL, with a reusable pre-registered protocol; (ii) the finding that loss aversion shifts it rightward, a direction whose sign we fix a priori and confirm across four environment–algorithm settings, with the metastable horizon slide generalizing likewise; (iii) a decomposition showing prospect theory's two components move it in opposite directions; and (iv) the result that a downstream loss-averse *minority* can set the boundary on a directed-flow substrate. Our novelty is narrow: Mengesha, Sun, and Roy (2025) show loss aversion moves a tipping point, but in evolutionary game theory over fixed strategies, and Lalmohammed et al. (2025) bring CPT into cooperative MARL without separating its components or measuring a boundary. Ours is the measured, CI'd boundary and the component decomposition against it, in gradient-trained neural policies.

Related Work

Coordination collapse and credit assignment. Cooperative MARL has been studied since independent Q-learning on shared tasks (Tan 1993; Buşoniu, Babuška, and De Schutter 2008), and deep variants inherit its pathologies—the lazy-agent and relative-overgeneralization failures in which agents abandon cooperation for a degenerate policy (Liu et al. 2023; Shi and Peng 2023; Tampuu et al. 2017). The standard remedy is centralized credit assignment under CTDE: a centralized critic or a factorized value function that decomposes team reward into per-agent contributions (Lowe et al. 2017; Yu et al. 2022; Sunehag et al. 2018; Rashid et al. 2018; Foerster et al. 2018; Son et al. 2019), benchmarked on suites such as SMAC (Samvelyan et al. 2019). Weng and Lee (2026) attribute the failure to representation degradation under decentralized exploration, and Akella (2025) show decentralized shaping cannot match centralized baselines. This literature establishes *that* centralization helps and *how* to implement it; none locates the threshold w_{crit} at which it becomes necessary, or attaches a confidence interval to it—the gap we close, using the centralization weight w as the dial rather than a fixed factorization.

Phase transitions in multi-agent learning. Treating collective learning as a dynamical system with critical points has a growing precedent. Yamaguchi (2025) construct a phase diagram for independent Q-learning over *environmental* axes (grid size, agent density), and Capannoli et al. (2026) prove a metastable partial-adoption boundary in network opinion dynamics under a fixed update rule. Closest in method, the deterministic-limit and mean-field programmes for multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019; Hussain, Belardinelli, and Leonardos 2025) derive Fokker–Planck descriptions of networked Q-learning and tie convergence to exploration and sparsity. We share their statistical-physics posture but differ in object: ours is a *reward-structure* threshold in gradient-trained cooperative policies, with confidence intervals, that a *behavioral* parameter then moves—and we apply the mean-field machinery as a known tool to fix the sign, not as the contribution itself.

Behavioral preferences in cooperative AI. A thin bridge connects human preference structure to collective learning dynamics: Hughes et al. (2018) show inequity aversion improves cooperation in social dilemmas, and the cooperative-AI agenda (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020) frames behavioral and safety considerations for multi-agent systems. On the risk-sensitive side, CVaR objectives optimize tail outcomes (Qiu et al. 2021; Shen et al. 2023; Prashanth 2014; Tamar, Glassner, and Mannor 2015)—distinct from loss aversion, which is a reference-point asymmetry rather than a tail reweighting. Prospect theory has been brought into single-agent RL for prediction and control (Prashanth et al. 2016), and its calibration concerns are well known (Rabin 2000; Tversky and Kahneman 1992b). In the multi-agent setting, Mengesha, Sun, and Roy (2025) prove loss aversion shifts a coordination tipping point, but in evolutionary game theory over fixed strategies; Lalmohammed et al. (2025) bring CPT into cooperative MARL but fold it into the actor–critic update, entangling its components. Our work extends this bridge to prospect-theoretic preferences in gradient-trained *cooperative* policies and, uniquely in this line, *decomposes* prospect theory's components against a measured boundary.

Preliminaries and Method

In cooperative MARL, N agents share an environment and maximize a joint objective under CTDE (Lowe et al. 2017); the reward-mixing weight $w \in [0, 1]$ interpolates purely local (w=0) and purely global (w=1) reward. As $w \to 0$ the cooperative signal is diluted and agents collapse to the safe lazy policy (Liu et al. 2023; Shi and Peng 2023). This w is our sole structural intervention; all else is held fixed, so any boundary shift is attributable to w and the behavioral parameters. Cumulative prospect theory (Kahneman and Tversky 1979; Tversky and Kahneman 1992a) gives the value function

$v(x) = x^\alpha$ for $x \geq 0$ and $-\lambda(-x)^\beta$ for $x < 0$, where $\alpha, \beta \in (0, 1]$ govern curvature and $\lambda > 1$ is loss aversion (α=β=1, λ=1 is linear utility). We apply it as a reward transform after blending,

$$r_i^{\mathrm{PT}} = v\big((1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}\big)\,, \qquad (1)$$

the decisive difference from Lalmohammed et al. (2025), who fold CPT *into* the actor–critic update, entangling curvature with loss aversion and confounding any shift with a change to the optimizer. Transforming only the scalar reward leaves the policy gradient identical across conditions, so a measured shift is behavioral and the two channels dial independently. We pre-registered the binary collapse label $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$ ($\bar{S}$ mean base-stock, $\bar{\pi}$ mean profit), freezing it in version control before any analysis (the commit hash is withheld for anonymity and released on acceptance); it is robust to ±20% threshold perturbation (max $\Delta w_{\mathrm{crit}} = 0.015$; Supplement). For each condition we fit $P(\text{collapse} \mid w) = \sigma(b_0 + b_1 w)$, estimate $w_{\mathrm{crit}} = -b_0/b_1$, and bootstrap its 95% CI; steep high-λ cells use Firth penalized regression, which agrees with the bootstrap throughout. The decomposition compares three conditions with identical environment and optimizer: **Null** (α=β=1, λ=1), **Curvature** (α=β=0.88, λ=1), and **Loss aversion** (α=β=0.88, $\lambda > 1$). The primary substrate is `NetworkManagement-v1` (Hubbs et al. 2020), a six-node multi-echelon inventory network that extends prior deep RL inventory work (Oroojlooyjadid et al. 2022; Gijsbrechts et al. 2022; Leluc et al. 2023); the transfer environments are MPE Cooperative Navigation (Lowe et al. 2017) and Level-Based Foraging (Christianos, Schäfer, and Albrecht 2020). Agents run PPO (Schulman et al. 2017) via Stable-Baselines3 (Raffin et al. 2021) and PettingZoo/SuperSuit (Terry et al. 2021; Terry, Black, and Hari 2020) (20 seeds/condition, 1,524 primary runs). Full setup and hyperparameters are in the Supplement.

The Boundary and Its Behavioral Shift

A sharp, metastable boundary. Collapse is a sharp logistic transition in w, not a gradual decay: at 300k steps $w_{\mathrm{crit}} = 0.666$ $[0.635, 0.694]$ (transition width $\Delta w = 0.199$, $p = 3.2 \times 10^{-7}$), with collapse fractions rising from near-zero to near-one within $\Delta w < 0.3$ (Figure 1a). The boundary is not a constant but a *metastable* threshold that slides left with the training horizon, and across four matched risk-neutral horizons ($w_{\mathrm{crit}} = 0.822, 0.666, 0.622, 0.573$ at 150k, 300k, 450k, 900k) the slide is *log-linear in* T: $w_{\mathrm{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027; all four CIs of width ≤0.04; Figure 1b). This log-linear form is the quantitative Kramers escape-time signature the mechanism predicts: each doubling of the budget buys a fixed ≈ 0.10 leftward shift in w, so the boundary keeps falling below the longest measured point (0.573 at 900k)—extrapolating the law to 2M steps gives $w_{\mathrm{crit}} \approx 0.44$. Collapse at intermediate w is therefore the metastable *approach* to the deterministic floor $w_{\mathrm{crit}}^0 = c/(c+g)$ the mechanism derives (§), escapable with more training rather than absorbing; we report the slide and its rate, and treat the floor's exact value as below the measured range rather than extrapolating the log-linear form to an (unphysical) zero crossing. The slide is not a supply-chain artifact either: under centralized-critic MAPPO it recurs in both transfer environments—on MPE w_{crit} falls $0.82 \to 0.68$ as the horizon grows 2M → 8M, and on Level-Based Foraging $0.33 \to 0.20$ over 1M → 4M, both with non-overlapping intervals. A basin-control experiment rules out initialization as the driver (Supplement).

Loss aversion raises the boundary; curvature lowers it. Against a single canonical null—$w_{\mathrm{crit}}^{\mathrm{null}} = 0.666$ $[0.635, 0.694]$, independently reproduced at 0.685 by a dedicated experiment (every other null-like number is a labeled offset, not a separate baseline)—we sweep $\lambda \in \{1, \ldots, 7\}$ (α=β=0.88, 300k). The boundary rises monotonically with loss aversion—$\{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$—then saturates near 0.85 with clear headroom to the w=1 ceiling, so the plateau is behavioral, not a ceiling artifact (Figure 2, Table 1); the endpoints are non-overlapping by 0.221. Curvature acts in the *opposite* direction, *lowering* w_{crit} to 0.586 (-0.100 $[-0.172, -0.033]$ vs. the null). That the two components of a single "risk-averse" psychology pull the boundary in opposite directions is the first of two surprises, and is visible only because there is a calibrated null to read each component against.

Two ways to count the loss-aversion effect. The magnitude depends on the baseline, and we report both explicitly to avoid ambiguity (Table 1). Measured *against the linear null* (0.685), the combined CPT shift is +0.089 $[0.041, 0.141]$ at λ=3 and +0.166 $[0.135, 0.202]$ at λ=7—but this nets loss aversion against curvature's -0.100. The *pure loss-aversion increment*, measured against the curvature baseline (λ=1, 0.586)—the correct isolate, since it holds curvature fixed—is larger: +0.188 at λ=3 and +0.265 at λ=7. The two are reconciled by the curvature offset ($-0.100 + 0.265 = 0.165 \approx 0.166$). Throughout, "the loss-aversion effect" refers to the increment over the curvature baseline; the net-vs-null figure is the deployment-relevant total. The full sweep is tracked by a one-parameter, theory-bounded affine law (§) to RMSE 0.014. A finer $w \in \{0.80, 0.90\}$ grid resolves the steep high-λ cells: the $\lambda \geq 5$ boundary is tight and cleanly monotone ($0.825, 0.835, 0.852$ at λ=5, 6, 7, CI widths $0.038, 0.033, 0.024$—about half those from a single w=0.85 cell).

A Mean-Field Mechanism Fixes the Sign

We give a mechanistic account of why the boundary exists and which way each component moves it. Following deterministic-limit and mean-field treatments of multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019; Hussain, Belardinelli, and Leonardos 2025), we model each Gaussian-policy

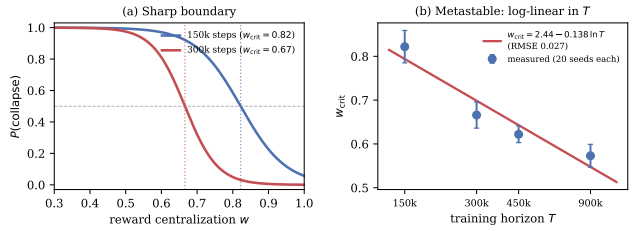


Figure 1: (a) Collapse is a sharp logistic transition in w; the boundary sharpens and slides left from 150k to 300k steps. (b) Across four matched risk-neutral horizons (20 seeds each, all CIs of width $\leq$0.04) w_{crit} falls *log-linearly* in T, $w_{\text{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027)—the Kramers metastable-escape signature the mechanism predicts.

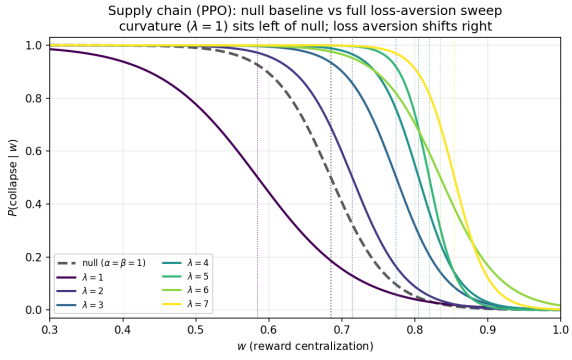


Figure 2: Behavioral decomposition and full loss-aversion sweep (supply chain, PPO). Curves are Firth-fitted $P(\text{collapse} \mid w)$; dotted lines mark w_{crit}. Curvature (λ=1) sits *left* of the null (easing coordination); loss aversion (λ=2, . . . , 7) shifts w_{crit} *right*, rising then saturating.

PPO agent's mean action as a Langevin variable; the machinery is standard and is in the Supplement. *We claim the mechanism at the level of sign and functional form, not magnitude.* Modelling the policy mean as $\mathrm{d}\mu_i = D(w)\,\mathrm{d}t + \sqrt{2\Theta(w)}\,\mathrm{d}W$, with drift D and a centralization-weighted diffusion Θ into which partner exploration enters, the replenishing regime makes an upstream node's inventory a local cost but a global benefit, giving cooperative drift $D(w) = \eta[wg - (1-w)c]$. Escape from the lazy basin is governed not by drift but by its ratio to gradient noise,

$$\mathrm{SNR}(w) = \frac{D(w)}{\sqrt{\Theta(w)}} \gtrsim \kappa_c, \tag{2}$$

and the crossing defines w_{crit}. Because escape is a Kramers first-passage problem (Kramers 1940) ($\tau \sim e^{\Delta U/\Theta}$)—the noise-induced analogue of a saddle-node bifurcation of the coordinating basin (Strogatz 1994)—collapse switches abruptly across w_{crit} (the sharp fits) and $w_{\text{crit}}(T)$ falls as T grows (the metastable slide),

Table 1: Three-way decomposition of w_{crit}. We report the effect both ways: net vs. the linear null (combines curvature and loss aversion), and the pure loss-aversion increment vs. the curvature baseline (0.586). "Sig." = difference CI excludes zero.

Condition	w_{crit}	vs. null	vs. curv.
Null (λ=1)	0.685	—	—
Curvature (λ=1)	0.586	$-0.100^{\checkmark}$	0
Loss av. λ=3	0.774	$+0.089^{\checkmark}$	+0.188
Loss av. λ=7	0.851	$+0.166^{\checkmark}$	+0.265

both predicted without fitting. We do *not* derive the numerical value of w_{crit}: c, g, σ, κ_c are calibrated, and our pre-registered direct drift probe was *inconclusive*—it reproduces the matched-horizon boundary in collapse fraction but does not localize w_{crit}, because survivor bias in the contested band confounds the conditioned drift (Supplement). The magnitude therefore remains calibrated; the mechanism's predictive content is the sign and the form.

The sign *propagates*—conditional on a stated payoff premise.

Proposition 1 (Sign propagation of the loss-aversion shift)**.** *Let the order parameter x evolve as $\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t$ with $\Sigma > 0$, and* assume the payoff premise $\partial_\lambda F < 0$ *on the escape interval—i.e. loss aversion scales only the loss domain and so inflates gradient noise. Then $\partial_\lambda w_{\text{crit}}^{\text{Emp}} > 0$.*

Proof sketch. The Kramers barrier $\Delta U = \int 2F/\Sigma\,\mathrm{d}z$ has $\partial_\lambda \Delta U < 0$ pointwise, since $\Sigma > 0$ and $\partial_\lambda F < 0$ throughout, independent of the sign of F; a lower barrier shortens escape ($\partial_\lambda \tau < 0$) while $\partial_w F > 0$ lengthens it. The implicit function theorem on $\tau(w, \lambda) = T$ gives $\mathrm{d}w_{\text{crit}}^{\text{Emp}}/\mathrm{d}\lambda = -\partial_\lambda \tau / \partial_w \tau > 0$. The magnitude depends on the intractable Σ; the direction does not. The same argument with curvature's drift signs reversed yields the opposite shift. □

We state this as a *proposition*, not a theorem, because its force is conditional: it assumes the payoff premise $\partial_\lambda F < 0$ rather than deriving it from first principles. We verify that premise directly on the substrate (a finite-difference probe confirms loss aversion inflates the loss-domain gradient variance; Supplement), so the result's content is that this *local* sign propagates to the *global* boundary regardless of the intractable noise Σ—which is exactly what makes the direction (but not the magnitude) predictable. The same drift analysis fixes the opposite sign of the two prospect-theory components without fitting: curvature ($\alpha, \beta < 1$) compresses both gains and losses, lowering the variance in Θ, raising the SNR, and so *lowering* w_{crit}, while loss aversion inflates Θ and *raises* it—matching the measured -0.100 and $+0.265$ (Table 1).

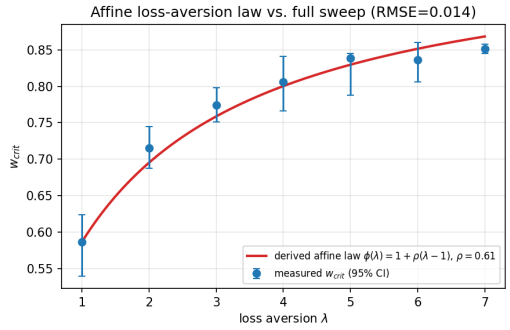


Figure 3: The derived affine law (3) (line, one free parameter $\rho = 0.61$ fixed on three points) against the measured $w_{\text{crit}}(\lambda)$ (points, bootstrap 95% CIs). It tracks the held-out points to RMSE 0.014, with a mild concave residual at the high-λ plateau.

A theory-bounded law, tested out of sample. The same loss-domain split fixes the functional *form*, not just the sign. Only the loss domain scales with λ, giving

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \qquad \phi(\lambda) = 1 + \rho(\lambda - 1), \quad (3)$$

with $\rho \in [0, 1]$ *bounded by the theory*—a genuine parametric constraint, not a free curve fit. We fix g/c from the λ=1 baseline and $\rho = 0.61$ from three calibration points ($\lambda \in \{1, 3, 7\}$), then test *out of sample* against the held-out four ($\lambda \in \{2, 4, 5, 6\}$): $w_{\text{crit}}^{\text{pred}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\}$ versus $w_{\text{crit}}^{\text{meas}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$, RMSE 0.014 across all seven points (Figure 3). The one systematic residual is a mild concave overshoot at the plateau, consistent with agents endogenously reducing loss exposure as λ grows. We report this derived, bounded law rather than a better-fitting power law because the point is mechanism, not goodness of fit—and we are explicit that it is *single-substrate*: it does not transfer to LBF (§). Table 2 states what the mechanism derives a priori, what is empirically confirmed, and what stays calibrated.

The Direction Generalizes; the Magnitudes Do Not

The mechanism commits in advance to a direction: loss aversion inflates gradient noise and so raises w_{crit}. If that is right rather than an artifact of one substrate or optimizer, the sign should survive changing both. We test on two further environments and a second algorithm, matching the learner to each environment's learnability (independent PPO where coordination is locally learnable; centralized-critic MAPPO (Yu et al. 2022) where the reward is shared-global). The direction is robust in every setting; the absolute level, the curvature sign, and the functional form are not (Figure 4, Table 4). On the *supply chain under MAPPO* (only the learner changes), w_{crit} rises 0.796 → 0.889 over λ=1 → 7, monotone from λ=2, endpoints non-overlapping; the level differs at low λ but converges with PPO by $\lambda \geq 4$. On *MPE* the loss-aversion excess is +0.158 [0.108, 0.212] at λ=7, but the curvature effect is null/opposite (+0.042, CI spanning zero), and on *LBF* curvature moves the boundary *right*—both confirming the curvature sign is substrate-specific. The supply chain's affine law misfits LBF by RMSE 0.15. Thus the *method*—a CI'd boundary that loss aversion shifts—generalizes across three environments and two algorithms, while the affine constants and the curvature sign do not.

Table 2: What the mechanism claims. The predictive content is sign and form; the magnitude is calibrated, and we make no closed-form claim for w_{crit}.

Derived (no fit)	opposite signs; monotone $w_{\text{crit}}(\lambda)$; affine ϕ with $\rho \in [0, 1]$; abrupt sharpness; w_{crit} falls with T
Confirmed	opposite signs, affine form; horizon slide (also MPE, LBF); exploration-noise dependence
Calibrated	g/c; loss-domain share ρ; noise scale
Not claimed	closed-form w_{crit}; magnitude on any axis

Two boundary conditions, both predicted. The mechanism makes the boundary contingent on two ingredients, and ablating each confirms it (Table 3). *Action representation.* The signal-to-noise crossing is a property of a continuous gradient, so destroying it should remove the boundary. Discretizing the base-stock control into K=11 bins and running value-based learners (QMIX, IQL) yields a flat, w-independent collapse rate with no fittable w_{crit}. Crucially, a single-variable control isolates the cause: PPO on the *same* discrete shell also goes flat, so it is coarse discretization—not the value-based paradigm—that erases the boundary, exactly as the continuous-gradient account predicts. (This is not a claim that discreteness per se removes the boundary: MPE is natively discrete and retains it.) *Credit assignment.* On a shared-reward task the boundary is measurable only with a learner that can coordinate at all: independent PPO collapses at *every* w on MPE—a coordination floor masking the behavioral boundary—while centralized-critic MAPPO recovers it. Neither is a negative result; each is a confirmed scope condition of the phenomenon.

A Minority Sets the Boundary

The decomposition treats the population as homogeneous, but deployed systems mix loss-averse and risk-neutral participants. Holding the substrate, PPO config, label, and grid identical to the main sweep, we ask two questions: how much bias at a critical node

Table 3: Action-representation ablation. The control (last row) changes only the action space while holding the learner at PPO, isolating coarse discretization as what erases the boundary.

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes
QMIX (centralized value)	discrete K=11	no
IQL (independent value)	discrete K=11	no
PPO (control)	discrete K=11	**no**

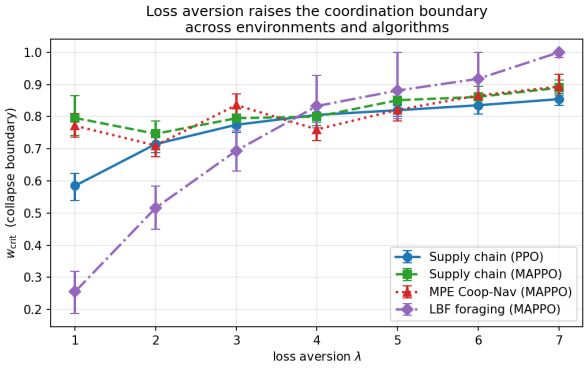


Figure 4: w_{crit} versus λ across four settings: supply chain under PPO and MAPPO, MPE under MAPPO, LBF under MAPPO (bootstrap 95% CIs). w_{crit} rises with λ in every setting; the *direction* generalizes, while levels (most at λ=1), curvature sign, and exact form are substrate-specific. LBF high-λ points saturate against the ceiling and are exploratory.

is needed to move the boundary, and which node's bias matters most (Figure 5).

A single node, dose for dose. Sweeping the most downstream trainable node's loss aversion $\lambda \in \{1, \ldots, 7\}$ with all others risk-neutral traces a clean, strictly monotone dose-response: w_{crit} rises 0.586 $\rightarrow$ 0.856 and converges on the *fully* loss-averse population value (0.851) by λ=7 (Figure 5b). One biased node, biased hard enough, imposes the centralization requirement of the entire network—the homogeneous result is the saturation of a continuous law, not a special case.

Position: proximity to demand, not chain index. Placing a single λ=7 node at each position, the effect is strongly position-dependent, but the governing variable is how directly a node's output reaches demand, not a one-dimensional chain index (Figure 5a). The two nodes feeding retail directly dominate (0.856 and 0.800, the strongest at the all-loss-averse boundary); among upstream suppliers, the one feeding *only* the dominant downstream node is nearly as strong (0.797), while those *splitting* output across both downstream nodes are weaker (0.726, 0.673). The network is a branching DAG, not a series chain, and w_{crit} tracks *path concentration*: a biased node matters more the more directly and exclusively its product flows to demand. That a behavioral minority moves a tipping point is known in evolutionary game theory (Mengesha, Sun, and Roy 2025); ours is the deep-MARL instantiation, with a topological signature absent from that account—audit the participant whose output most directly reaches demand, not the average or merely "upstream" one. The result is single-substrate *by construction*: "which node sets the boundary" is well-posed only under a directed resource flow, which MPE and LBF lack (Supplement). We frame its reach as a *falsifiable* prediction: in any directed-flow cooperative domain, the boundary is set by the biased participant of highest path concentration to demand.

Table 4: Generalization of the loss-aversion *direction*. The sign holds everywhere; only the supply chain (PPO) yields the affine law. Detail in the Supplement.

Setting (algo.)	Loss-av. effect	Affine law?
Supply (PPO)	+0.166 (λ=7)	yes (0.014)
Supply (MAPPO)	0.80$\rightarrow$0.89	—
MPE (MAPPO)	+0.158 (λ=7)	no
LBF (MAPPO)	0.26$\rightarrow$0.83	no (0.15)

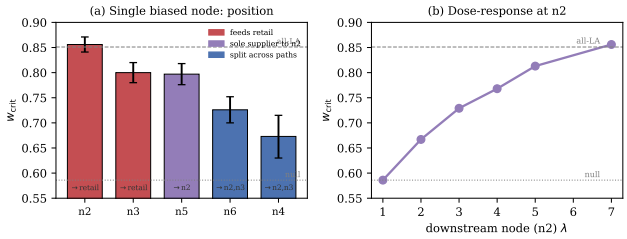


Figure 5: A minority sets the boundary. (a) A single λ=7 node by position: nodes feeding retail directly (red) dominate; among upstream suppliers, the one feeding *only* the dominant downstream node (purple) nearly matches them, while those *splitting* output across paths (blue) are weaker—w_{crit} tracks path concentration to demand, not chain index. (b) Dose-response: sweeping the downstream node's λ traces a strictly monotone rise to the homogeneous all-loss-averse value (0.851, dashed).

Discussion

That probability weighting and loss aversion exert opposite effects on the boundary is, to our knowledge, novel; prior work applies prospect theory as a unified model without decomposing it (Lalmohammed et al. 2025; Ghaemi et al. 2024). The operational reading is concrete: a system of λ=7 agents needs $w > 0.851$ for reliable coordination against $w > 0.685$ for risk-neutral agents—a +0.166 deployment margin that scales with λ, so domains nearer $\lambda \approx 2$–3 incur a smaller but positive

cost. Where centralization is capped by privacy or communication cost, the boundary reframes the problem, exposing λ—the agents' effective loss aversion, shaped by the training objective—as a second lever. Our phase-diagram framing sits within a recognized line of multiagent phase-transition results (Yamaguchi 2025; Capannoli et al. 2026) and the broader project of cooperative AI (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020); its distinctive contribution is a boundary set by the policy-gradient signal-to-noise ratio rather than a fixed update rule.

What a practitioner does with this. The boundary is a design quantity. A team deploying cooperative agents can *estimate* its own w_{crit} with the pre-registered protocol (turning "how much should we centralize?" into a measured target with error bars), *budget* for the loss-aversion margin on top of the risk-neutral requirement rather than discovering collapse in production, and—where w is capped by privacy or communication cost—treat λ as the second lever the decomposition exposes, lowering agents' *effective* loss aversion through the training objective instead of raising w. The minority result adds a targeting rule: audit the biased participant whose output most directly reaches demand, not the average agent. The relevant λ is itself set by how an agent was trained—preference-aligned objectives are prospect-theoretic in form (Ethayarajh et al. 2024; Tversky and Kahneman 1992b)—so estimating it from behavior, against the calibration debates around prospect theory (Rabin 2000), ties the behavioral axis to a measurable property of the training pipeline. Laboratory loss aversion is $\lambda \approx 2.25$, so the deployment-relevant margins sit near our λ=3 figure.

Scope and Future Work. Three boundaries of the present results each point to a concrete next step. *(i) From one substrate to many.* The *quantitative* core—the $w_{\text{crit}}(\lambda)$ law and the cascade—is established on a single directed-flow substrate by construction; we replicate the boundary, the loss-aversion direction, and the metastable slide across environments and a second algorithm, but the precise magnitudes are a property of one simulator and one compute budget, and the law itself does not transfer to LBF. Replicating the cascade on a second directed-flow domain (e.g. congested traffic routing or staged resource allocation) would convert "single-substrate by construction" into a tested generalization. *(ii) From calibrated to predictive.* The mechanism fixes sign and form but not magnitude: our one direct probe of the policy-gradient drift was *inconclusive*, reproducing the boundary in collapse fraction but failing to localize w_{crit} because the policy initializes above the cooperative optimum, so collapses cross the contested band too fast to sample and bias the conditioned drift toward survivors. A probe that initializes near the cooperative optimum, or measures a basin-flow statistic rather than the conditioned drift, would turn the magnitude from a fit into a prediction and discharge the payoff premise that Proposition 1 currently assumes—the single change that would most strengthen the account. *(iii) From bracket to calibration.* We sweep λ to a high-stakes upper bound of 7, beyond the typical human value ($\lambda \approx 2.25$, with deployment-relevant margins near $\lambda \approx 3$); estimating the effective λ of preference-aligned agents from behavior would tie the behavioral axis to a measurable property of the training pipeline. None of these undercut the central, replicated claim—loss aversion raises the coordination requirement—but each marks where measurement currently ends and prediction begins.

Conclusion

We presented the first confidence-intervalled phase diagram of cooperative-MARL coordination collapse, characterizing w_{crit} as a function of reward structure, training horizon, and prospect-theoretic loss aversion. Loss aversion impedes coordination—raising the requirement, a direction we predict a priori and confirm across four environment–algorithm settings, with the metastable slide generalizing likewise. Two further results sharpen the picture: prospect theory's two components act in opposite directions, and a downstream loss-averse minority can set the boundary on a directed-flow substrate, framed as a falsifiable prediction. Beyond the findings, the pre-registered phase-diagram protocol—a frozen collapse label, a logistic fit over a small w-sweep, and a bootstrap interval on w_{crit}—is a reusable instrument: any cooperative-MARL study can report *where* its system collapses, with error bars, rather than asserting that centralization helps.

The clearest paths forward—a non-survivor-biased probe that upgrades the mechanism from calibrated to predictive, and a second directed-flow substrate that tests the cascade's reach—are set out above. More broadly, a measured collapse boundary turns "how should we incentivize a team of human-shaped learning agents?" into a question with a number, an error bar, and a named participant who sets it.

Ethical Statement

This work studies when cooperative multi-agent systems fail to coordinate, in a simulated inventory substrate with no human subjects or personal data. Its intended use is defensive: the boundary is a safety margin that helps practitioners avoid silent coordination collapse before deployment, and the minority result flags which participant most needs scrutiny. Two cautions follow. Raising the centralization dial w trades agent autonomy and local information for collective reliability, a trade with governance and privacy implications a technical threshold does not settle; and the behavioral axis we expose could be tuned to manipulate a system's coordination requirements rather than safeguard them—we report the decomposition and cascade precisely so such effects stay auditable. Our quantitative claims are scoped to one simulated substrate and should be re-measured before transfer to real deployments.

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